Confidence isn't intelligence

Over the past two years, I've lost count of how many times someone has watched an AI explain a complicated physics concept in flawless English, and declared how LLMs would be beating scientists at their own game soon enough. I mean OpenAI even has a research agent within ChatGPT which can already crack PhD level exams like its just another tuesday for it!

I get the impulse to jump to conclusions, I've felt it too. When you spend your life around tech, there's a particular dopamine hit that comes from watching a system manage to do something in a wink which used to require humans years of training. Especially when an AI does a particular action rather confidently and fluently.

But confidence, as every working journalist and researcher knows, is cheap.

A recent research paper on evaluating large language models in scientific discovery landed with me precisely because it punctured that illusion without theatrics. No doom, no hype. Just a careful attempt to answer an awkward question that I've often wondered about. When the likes of ChatGPT speak sophisticatedly about science concepts, does it mean they can do cutting-edge science research, or are they just very good at talking like people who do?

Because most benchmarks we use to judge "AI for science" look suspiciously like exams. Multiple-choice questions. Short answers. They are structured problems with known solutions. They reward recall and composure – just like students cramming their entire syllabus a night before finals. They do not reward the thing that defines real scientific research – like being wrong, noticing mistakes, and changing course.

Anyone who's ever sat through a failed experiment knows that discovery is less eureka and more paperwork. Hypotheses don't just appear, they're tracked, revised, and often discarded. Data arrives noisy and uncooperative. Progress is measured in slow, repetitive steps that don't work.

The research study suggests that when models are pushed into that messier loop – propose a hypothesis, design a test, interpret results, revise their beliefs – they start to crack. Not catastrophically, but just enough to reveal the gaps that are still there.

They cling to ideas a little too long, ignoring inconvenient results that just don't fit into their worldview. They prefer explanations that sound coherent over ones that are backed with concrete evidence.

I've seen this play out repeatedly in my own use of many such AI tools. They're fantastic accelerants at the beginning of a process – brainstorming angles, summarising literature, suggesting new directions I might not even have considered. But if left unchecked, they also have a habit of rewarding my own laziness. If I lower my standards, they just happily meet me there lazily.

And that's the uncomfortable part of this moment heading into 2026, I feel. The risk isn't that AI is overhyped, but that our expectations have gotten way too low. We don't just mistake fluency as intelligence, we celebrate it. We keep mistaking plausible answers for water-tight, factual ones.

Of course, the road to hell is paved with good intentions. If an LLM can be made to change its thought process by rogue elements through a fake scientific loop, it will set alarm bells ringing from an AI ethics and safety perspective. But maybe there's a middle ground that balances risks to allow for true ingenuity sparks to fly.

This study doesn't argue against AI in science. It argues against illusion. It reminds us that discovery is a system, a painstaking process that's longer than a confident looking statement. If we want machines that genuinely help us find new truths, we need to demand more than articulate outputs. Most of all, we need to raise our own bar.

AI will constantly keep getting better. That part is inevitable. What remains to be seen is whether we grow more discerning alongside it, or do we continue to settle for being impressed by the sound of intelligence instead of its true substance. d

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